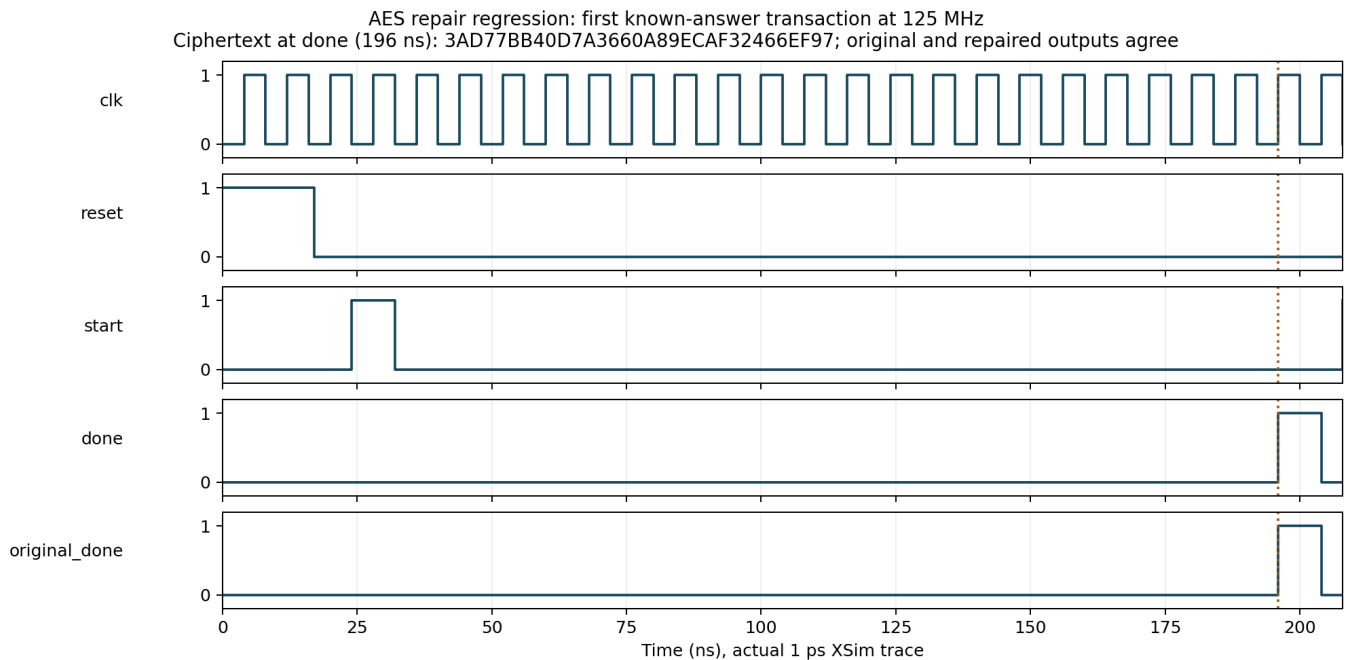


AES timing repair - simulation and evidence review

Beyond Boolean: Shanshank Pulipati (25BEC0573); Kartik Yadav (25BEC0087). Vivado XSim 2025.1.1, 10 October 2026. DUT: Timing_Repair/RTL/aes_core.v; comparison: Testbench/aes_core_original.v.

The testbench checks 104 deterministic known/reference vectors at 125 MHz and compares the original and repaired ciphertext/done each cycle. All reference ciphertexts were recomputed independently with PyCryptodome AES-ECB. It also checks input capture, reset during encryption, restart, held start, 21-cycle completion and retention.



Actual VCD rendered as a plot. Source VCD and vendor console log remain beside this PDF; no schematic or hardware image is generated by this plot.

PASS: 104 independent AES vectors at 125 MHz simulation clock
PASS: 21-cycle completion, one-cycle done, retained ciphertext
PASS: input capture, mid-encryption reset, restart, held start
PASS: cycle-by-cycle original/repaired equivalence throughout

The original one-vector 100 MHz simulation remains in Simulation/simulation_report.pdf. Behavioral simulation is distinct from static timing and board execution. Neither repaired pair has been physically verified during this review.